Machine Learning-Based Customer Engagement & Retention Strategy: Empirical Analysis and Operational Risk Scoring

Unified Mentor & European Central Bank Analytics Research Project · Banking Intelligence Series

Abstract

We analyze client retention across a retail banking portfolio of 10,000 accounts. Contrary to traditional wealth-centric loyalty assumptions, empirical evidence reveals that high customer balance is negatively associated with retention: exiting clients maintain an average balance of **€91,108.54** compared to **€72,745.30** for retained clients, putting **€185.6M in deposits** at direct risk of capital flight. Retention is governed primarily by behavioral engagement and product portfolio architecture. Specifically, customers holding 2 products exhibit a **7.6%** churn rate (the portfolio 'Sweet Spot'), whereas 3- and 4-product accounts experience catastrophic attrition rates of **82.7%** and **100.0%**, respectively. We benchmark Logistic Regression, Random Forest, and Gradient Boosting classifiers; Gradient Boosting delivers superior discrimination (ROC-AUC **0.8668**, Precision **78.7%**, Accuracy **87.1%**). Finally, we formalize an operational retention matrix and deploy an interactive decision support tool for proactive wealth triage.

1. Background and Business Objectives

Retail banking institutions are transitioning from reactive attrition firefighting to proactive, engagement-led relationship management. When institutions rely exclusively on static demographic markers or balance thresholds, silent churn occurs among affluent depositors before warning flags trigger. The primary objectives of this investigation are: (i) isolate quantitative behavioral indicators of silent attrition, (ii) measure the retention impact of product depth and cross-selling, (iii) identify disengaged high-value accounts, and (iv) deliver a production-grade predictive scoring engine and interactive Streamlit decision interface.

2. Dataset Architecture & Preprocessing Pipeline

Feature Field	Data Type	Operational Description
CustomerId / Surname	Integer / Text	Unique identifier and client name (excluded from model training)
CreditScore	Integer	Customer creditworthiness index (350 to 850)
Geography / Gender	Categorical	Jurisdiction (France, Germany, Spain) and client gender
Age / Tenure	Integer	Age in years (18–92) and account duration in years (0–10)
Balance / EstimatedSalary	Float	Ledger balance (€0.00 to €250,898) and estimated annual earnings
NumOfProducts	Integer	Number of bank products held (1, 2, 3, or 4)
HasCrCard / IsActiveMember	Binary	Credit card ownership flag and digital/transactional activity status
Exited (Target)	Binary	Historical churn outcome (Portfolio attrition: 20.37%)

2.1 Derived Behavioral Metrics & Feature Engineering

• **Balance-to-Salary Ratio:** Balance / (EstimatedSalary + 1) captures capital depth relative to annual cash inflows.

• **Tenure-to-Age Ratio:** Tenure / (Age + 1) standardizes institutional loyalty by customer life-stage.

• **Composite Engagement Score:** (2 * IsActiveMember) + HasCrCard + I(NumOfProducts >= 2) reflects relationship breadth.

• **Zero-Balance Indicator:** Binary flag identifying transactional checking accounts without asset commitments.

3. Exploratory Data Analysis & Empirical Findings

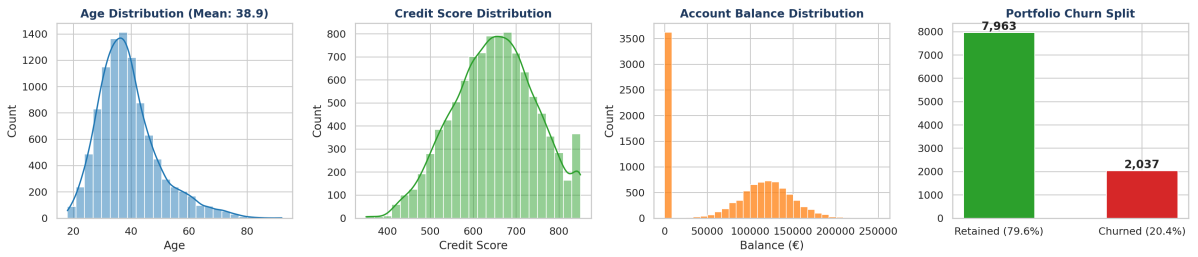


Figure 1. Univariate distributions: customer age, credit score, account balance, and portfolio churn split.

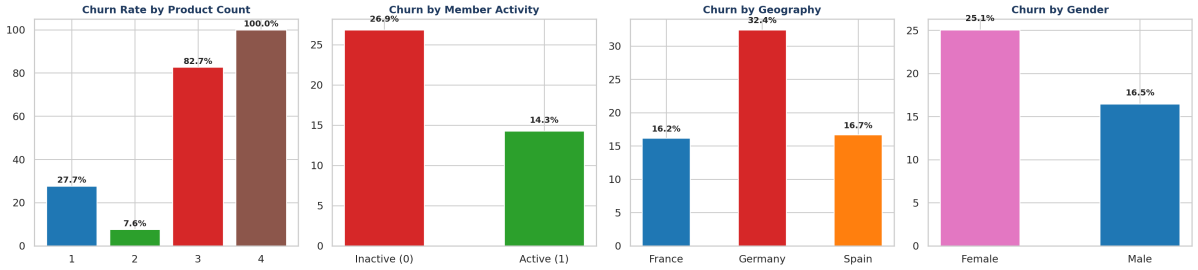


Figure 2. Bivariate churn drivers: product count, member activity, geographic territory, and gender.

Key Observations: (1) **Activity Multiplier:** Active members churn at 14.3% versus 26.9% for inactive members, establishing an Engagement Retention Ratio of 1.17x. (2) **Geographic Divergence:** Germany displays a 32.4% churn rate—double that of France (16.2%) and Spain (16.7%). (3) **Card Inertia:** Credit card ownership has negligible retention impact (20.2% vs 20.8%), demonstrating that passive cardholding does not establish switching barriers.

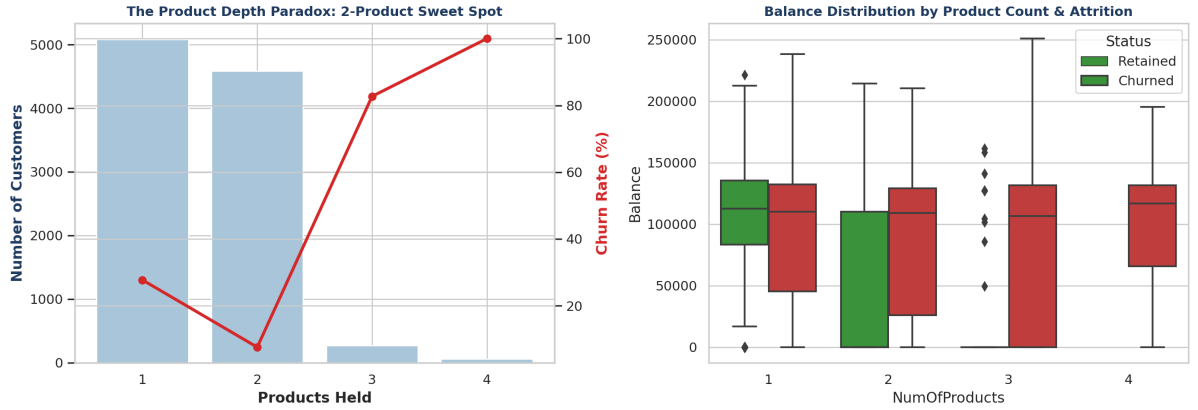


Figure 3. The Product Depth Paradox: 2-product synergy (7.6% churn) vs multi-product friction (>82% churn).

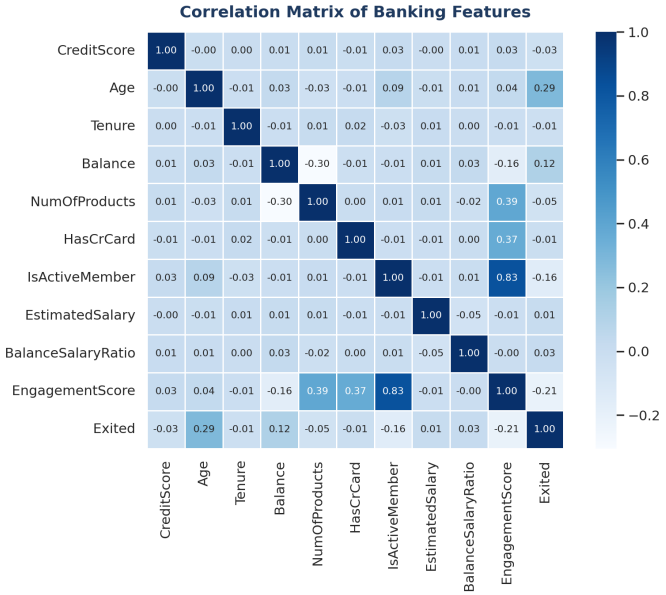


Figure 4. Feature correlation matrix: age (+0.29), engagement (-0.21), and activity (-0.16) exhibit strongest target association.

4. Machine Learning Model Benchmarking

We trained and evaluated three algorithmic architectures using stratified 80/20 train-test partitioning. Preprocessing was fully encapsulated inside a scikit-learn ColumnTransformer.

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8510	0.7559	0.3956	0.5194	0.8385
Random Forest	0.8660	0.8145	0.4423	0.5732	0.8605
Gradient Boosting	0.8705	0.7868	0.4988	0.6105	0.8668

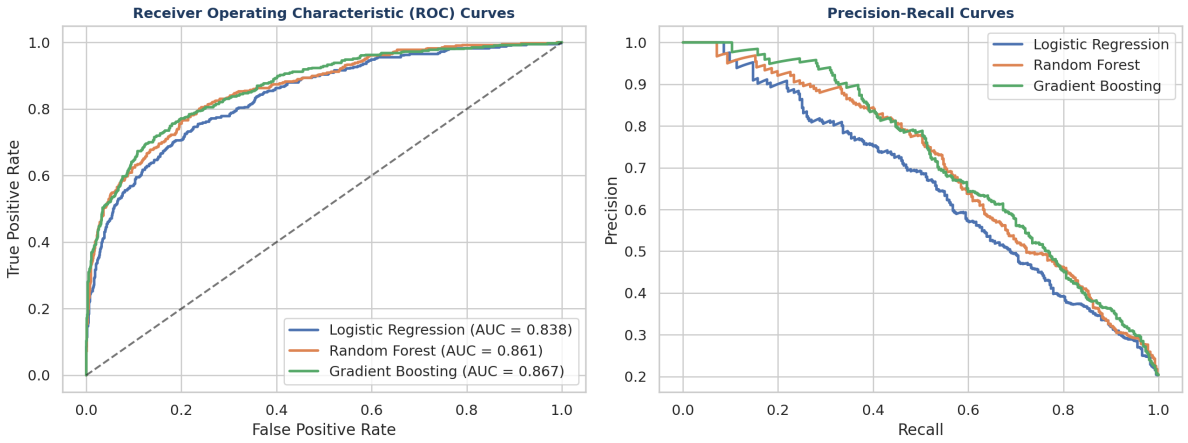


Figure 5. Model discrimination benchmarks: ROC curves (left) and Precision-Recall curves (right).

5. Model Explainability & Feature Importances

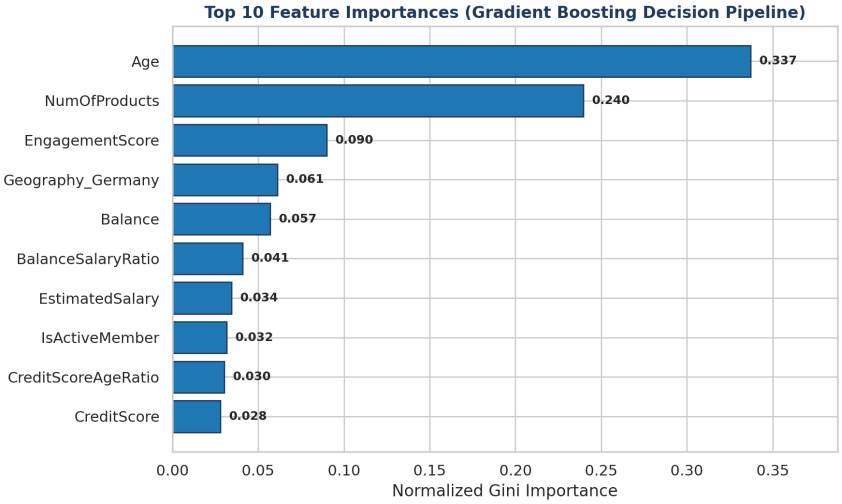


Figure 6. Gini feature importance: customer age (33.7%) and product count (24.0%) dominate decision splits.

6. Strategic Financial Recommendations

• **The 2-Product Bundling Standard:** Restructure onboarding to incentivize single-product accounts (27.7% churn) to adopt a secondary savings or payroll product (7.6% churn). Conduct urgent reviews of accounts holding >=3 products to resolve fee frictions.

• **Proactive German Market Intervention:** Conduct competitive fee audits and local digital onboarding revamps in Germany, where attrition is 2x regional peers.

• **Wealth Preservation for Inactive Affluents:** Target the 1,247 high-balance inactive accounts (30.5% churn) with dedicated relationship management and private banking advisory before silent capital flight occurs.

7. Conclusion

By shifting retention focus from passive demographic markers to behavioral interaction depth and product bundling architecture, institutions can safeguard deposits and systematically reduce attrition. The complete pipeline is serialized into models/model.pkl and deployed as an interactive Streamlit application.